



WEBIGEEKS

Full Stack / MERN Stack Development

Become a Job-Ready Full Stack Developer



Duration

7 Months – Intensive, Industry-Aligned Curriculum



Mode

Online & Offline – Flexible Learning Options



Contact

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COURSE OVERVIEW

Your Journey to Full Stack Mastery

The WebiGeeks Full Stack / MERN Stack Development programme is a comprehensive, 7-month training designed to transform complete beginners and career-switchers into confident, job-ready developers. Every module is meticulously crafted to mirror real industry workflows – from writing your first line of HTML to deploying full-stack applications on the cloud.

01

Web Foundations

HTML5, CSS3, Bootstrap – build pixel-perfect, responsive interfaces from day one.

02

JavaScript & DOM Mastery

Core JS, ES6+, DOM manipulation, jQuery, and asynchronous programming.

03

Backend with Node.js & Express

Server-side development, REST APIs, middleware, authentication, and databases.

04

React.js & Full Stack Integration

Modern frontend with React, connecting to backends, deployment, and live projects.

HTML5 — The Backbone of the Web

Every great web application begins with solid HTML. This foundational module ensures you develop a thorough understanding of how web pages are structured, semantically written, and made accessible to all users. You will learn to write clean, standards-compliant markup that search engines love and browsers render flawlessly.

Core HTML Concepts

- Introduction to HTML & how browsers parse markup
- HTML document structure: DOCTYPE, html, head, body
- Headings (h1-h6), paragraphs, line breaks
- Links (anchor tags), absolute vs. relative URLs
- Images, audio, video – embedding media
- Lists: ordered, unordered, and nested lists
- Tables: tr, td, th, colspan, rowspan

Advanced HTML & Forms

- HTML5 Semantic elements: header, footer, nav, section, article, aside
- Forms: input types, labels, placeholders, required attributes
- Form validation (HTML5 built-in)
- Iframes, meta tags, SEO basics
- HTML entities and special characters
- Accessibility best practices (ARIA roles)
- HTML comments and code organisation

CSS3 — Design, Style & Responsiveness

CSS transforms raw HTML into beautiful, engaging interfaces. This module takes you from basic styling to advanced layout techniques, animations, and responsive design. You will gain the confidence to recreate any design mockup and build visually stunning, mobile-first websites.

CSS Foundations

- Selectors, specificity & the cascade
- Box model: margin, padding, border
- Typography: fonts, sizes, line-height
- Colors, backgrounds, gradients
- Display: block, inline, inline-block

Layouts & Positioning

- Flexbox – complete mastery
- CSS Grid – 2D layout system
- Position: static, relative, absolute, fixed, sticky
- Z-index and stacking contexts
- CSS variables (custom properties)

Responsive & Advanced

- Media queries & breakpoints
- Mobile-first design strategy
- CSS transitions & keyframe animations
- Pseudo-classes & pseudo-elements
- CSS transforms (translate, rotate, scale)

Bootstrap 5 — Rapid UI Development

Bootstrap is the world's most popular CSS framework, enabling developers to build fully responsive, professional-grade interfaces in a fraction of the time. You will learn to harness Bootstrap's powerful grid system, component library, and utility classes to rapidly prototype and deliver polished UI layouts that work seamlessly across all devices.

Bootstrap Core

- Installing Bootstrap (CDN & npm)
- The 12-column Grid System
- Containers, rows, and columns
- Responsive breakpoints (xs, sm, md, lg, xl, xxl)
- Typography & colour utility classes
- Spacing utilities (margin, padding shorthand)
- Display and flexbox utilities

Bootstrap Components

- Navbar – responsive navigation
- Cards, buttons, badges, alerts
- Modals, tooltips, popovers
- Carousel (image sliders)
- Forms & form validation UI
- Tables, pagination, breadcrumbs
- Building complete responsive pages with Bootstrap

JavaScript — The Language of the Web

JavaScript is the engine that drives every modern web application. This module delivers a deep, thorough understanding of the language – from fundamental syntax to advanced concepts like closures, promises, and asynchronous programming. You will write JavaScript that is efficient, readable, and production-ready.

JS Fundamentals

- Variables: var, let, const
- Data types, operators, type coercion
- Conditionals: if/else, switch, ternary
- Loops: for, while, for...of, for...in
- Functions: declarations, expressions, arrow functions
- Scope, hoisting, closures

Objects & Arrays

- Arrays: map, filter, reduce, forEach, find
- Objects: creation, destructuring, spread/rest
- JSON: parse and stringify
- Template literals, optional chaining
- ES6+ features: classes, modules (import/export)
- Error handling: try/catch/finally

Async JavaScript

- Callbacks and callback hell
- Promises: .then(), .catch(), chaining
- Async/Await syntax
- Fetch API & making HTTP requests
- Working with third-party APIs
- Event loop, call stack, and microtasks

DOM Manipulation & jQuery

The Document Object Model (DOM) is the bridge between your JavaScript logic and the visual interface. Mastering DOM manipulation gives you the power to build dynamic, interactive web experiences. jQuery, while a mature library, remains widely used across existing codebases and teaches critical programming patterns that every developer must understand.

DOM Manipulation (Vanilla JS)

- Understanding the DOM tree structure
- Selecting elements: `getElementById`, `querySelector`, `querySelectorAll`
- Modifying text content and `innerHTML`
- Changing styles and CSS classes dynamically
- Creating, appending, removing DOM elements
- Event listeners: `click`, `submit`, `keydown`, `mouseover`
- Event delegation and event bubbling/capturing
- Form handling and input validation with JS
- Local Storage & Session Storage

jQuery

- jQuery installation and the `$` selector
- DOM traversal: `parent`, `children`, `siblings`, `find`
- jQuery events: `.on()`, `.click()`, `.hover()`
- jQuery animations: `show/hide`, `fadeIn/fadeOut`, `slideToggle`
- AJAX with jQuery: `$.ajax()`, `$.get()`, `$.post()`
- Chaining jQuery methods
- jQuery plugins and their integration
- DOM manipulation with jQuery vs. vanilla JS – when to use which

Node.js — Server-Side JavaScript

Node.js revolutionised web development by bringing JavaScript to the server. With Node.js, you can build fast, scalable, and event-driven backend applications using the same language you write on the frontend. This module covers Node.js from the ground up, giving you the confidence to architect robust server-side systems.

Node.js Foundations

- What is Node.js? Event-driven, non-blocking I/O model
- Installing Node.js and npm
- Node.js REPL and running .js files
- Global objects: `__dirname`, `__filename`, `process`
- Node.js module system (CommonJS: `require/module.exports`)

Core Built-in Modules

- `fs` module: reading/writing files, streams
- `path` module: file path management
- `os` module: operating system information
- `http` module: creating a basic HTTP server
- `events` module: EventEmitter pattern

npm Ecosystem

- `package.json` and `package-lock.json`
- Installing, updating, removing packages
- Local vs. global packages
- Scripts in `package.json` (`start`, `dev`, `test`)
- `nodemon` for development workflow

Express.js — Building Web Servers & APIs

Express.js is the de-facto web framework for Node.js, used by thousands of companies worldwide to build everything from simple REST APIs to complex enterprise backends. This module teaches you to build production-quality servers with clean architecture, proper routing, middleware pipelines, and template-based rendering.

Express Core

- Setting up an Express application
- Routing: GET, POST, PUT, DELETE
- Route parameters and query strings
- Request & Response objects (req, res)
- Static file serving with `express.static()`
- Sending JSON responses vs. HTML responses
- Environment variables with `dotenv`

Middleware & Templates

- What is middleware? The middleware pipeline
- Built-in middleware: `express.json()`, `express.urlencoded()`
- Third-party middleware: `morgan`, `cors`, `helmet`
- Writing custom middleware functions
- Error-handling middleware
- EJS templating: rendering dynamic HTML
- `Partials`, `layouts`, and template inheritance in EJS

Git & GitHub — Version Control Essentials

Version control is a non-negotiable skill in every professional development environment. Git and GitHub are the industry standard tools for tracking code changes, collaborating with teams, and contributing to open-source projects. This module builds your Git confidence from beginner to industry-ready, covering workflows that mirror real company practices.

Git Fundamentals

git init, git clone, git status, git add, git commit, git log, git diff – understanding the working tree, staging area, and commit history. Writing meaningful commit messages.

Branching & Merging

Creating and switching branches, git merge, merge conflicts and how to resolve them, git rebase, git stash, and cherry-pick. Feature branch workflow for team collaboration.

Remote Repositories & GitHub

git push, git pull, git fetch, git remote. Creating GitHub repositories, pull requests (PRs), code reviews, forking, and contributing to open-source projects. GitHub Pages for deployment.

Advanced Git Workflows

.gitignore configuration, GitHub Actions basics, SSH key setup, git reset vs. git revert, tags and releases, and team collaboration using GitHub Projects and Issues.

Databases — SQL, MongoDB & Mongoose

A full-stack developer must be proficient with both relational and NoSQL databases. This dual-database module teaches you to design schemas, write queries, and integrate databases seamlessly into your Node.js applications – giving you the flexibility to choose the right tool for any project.

SQL (Relational Databases)

- Relational database concepts: tables, rows, columns, keys
- SQL syntax: SELECT, INSERT, UPDATE, DELETE
- WHERE, ORDER BY, GROUP BY, HAVING clauses
- JOINS: INNER, LEFT, RIGHT, FULL OUTER
- Aggregate functions: COUNT, SUM, AVG, MAX, MIN
- Primary keys, foreign keys, indexes
- Database normalisation (1NF, 2NF, 3NF)
- Connecting SQL databases to Node.js applications

MongoDB & Mongoose

- NoSQL concepts: documents, collections, flexibility
- MongoDB installation & MongoDB Atlas (cloud)
- CRUD operations with MongoDB shell
- Mongoose ODM: connecting, defining schemas & models
- Schema types, validation, default values
- Mongoose queries: find, findById, findOne, findByIdAndUpdate, findByIdAndDelete
- Population (joins in Mongoose), references
- Indexes and query optimisation

REST APIs & Authentication

Building secure, well-structured APIs is at the heart of every full-stack application. This combined module teaches you to design and implement RESTful APIs following industry best practices, and to secure them with robust authentication and authorisation mechanisms – essential skills for any professional backend developer.

REST API Design

- REST principles: statelessness, resources, HTTP verbs
- API endpoint structure & naming conventions
- HTTP status codes (200, 201, 400, 401, 403, 404, 500)
- Request/response body with JSON
- Postman for API testing and documentation
- CRUD API with Express + MongoDB
- Error handling and validation middleware

Authentication & Authorisation

- Session-based vs. token-based authentication
- JWT (JSON Web Tokens): structure, signing, verifying
- Passport.js: local strategy, OAuth (Google, GitHub)
- Password hashing with bcrypt
- Protected routes and middleware guards
- Role-based access control (RBAC)
- Express-session and cookie management

Security Best Practices

- CORS configuration
- Helmet.js for HTTP security headers
- Rate limiting to prevent brute-force attacks
- Input sanitisation and injection prevention
- Storing secrets with environment variables
- HTTPS and SSL basics
- Refresh tokens and token expiry strategies

React.js — Modern Frontend Development

React is the world's most popular frontend library, used by Facebook, Netflix, Airbnb, and thousands of companies globally. This comprehensive React module takes you from your very first component to building complex, state-managed, API-connected applications – the pinnacle of the WebiGeeks curriculum.

React Fundamentals

- What is React? Virtual DOM & reconciliation
- Create React App / Vite setup
- JSX syntax and rules
- Functional components vs. class components
- Props: passing data between components
- Component composition and reusability

State & Hooks

- `useState`: managing local component state
- `useEffect`: side effects, API calls, cleanup
- `useContext`: global state without prop drilling
- `useRef`: DOM references and mutable values
- `useMemo` & `useCallback` for performance
- Custom hooks: creating reusable logic

Advanced React

- React Router v6: client-side navigation
- Dynamic routes and URL parameters
- Axios / Fetch for API integration
- Forms and controlled components
- Context API for state management
- Introduction to global state patterns

Redux & Redux Toolkit

- What is Redux? Flux architecture & state flow
- Redux core: store, actions, reducers
- Redux Toolkit (RTK) – the modern standard
- `createSlice`, `createAsyncThunk`
- RTK Query for data fetching & caching
- Integrating Redux with React (`react-redux`)
- `useSelector` & `useDispatch` hooks
- DevTools for debugging state

TypeScript — Scalable, Type-Safe JavaScript

TypeScript is the industry standard for large-scale JavaScript development, adopted by Google, Microsoft, Airbnb, and virtually every major tech company. This module equips you with the type-safety skills that make you a more reliable, hireable developer.

TypeScript Fundamentals

- What is TypeScript? JS superset & compilation
- Setting up TypeScript with tsconfig.json
- Type annotations: string, number, boolean, any
- Arrays, tuples, and enums
- Type inference and type aliases
- Union types and intersection types

Functions & Interfaces

- Typing function parameters and return types
- Optional and default parameters
- Interfaces vs. type aliases
- Extending interfaces
- Readonly and optional properties
- Generics: writing reusable typed code

TypeScript with React

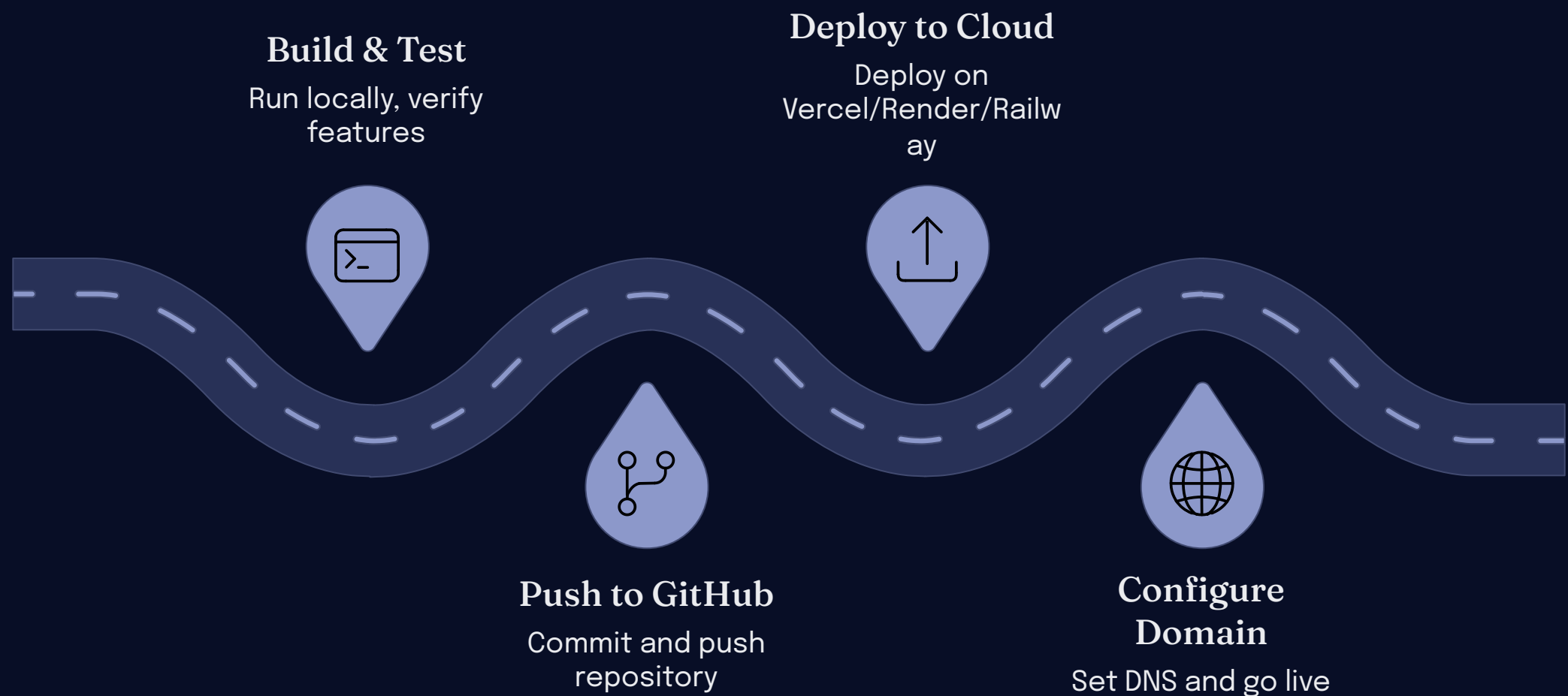
- Setting up a React + TypeScript project (Vite)
- Typing props with interfaces
- Typing useState, useRef, useEffect
- Typing event handlers and form elements
- Typing API responses and async functions
- React.FC vs. typed function components

Advanced TypeScript

- Utility types: Partial, Required, Pick, Omit
- Type narrowing and type guards
- Discriminated unions
- Typing Redux state and actions with RTK
- TypeScript in Express.js backends
- Migrating a JavaScript project to TypeScript

Deployment — Taking Your App Live

Building an application is only half the journey – deploying it to the world is where your work truly comes alive. This module teaches you the complete deployment workflow for both frontend and full-stack applications, using modern cloud platforms that are the industry standard for production deployments.



Master the end-to-end deployment pipeline that real companies use every day.

Frontend Deployment

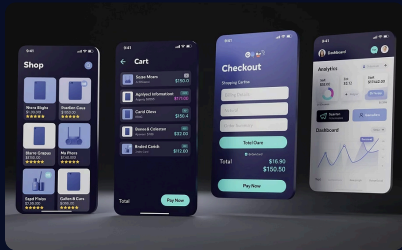
- Building React apps for production (npm run build)
- Deploying to Vercel – automated GitHub integration
- Netlify deployment and continuous deployment (CD)
- GitHub Pages for static sites
- Environment variables in production
- Custom domain configuration and DNS basics

Backend & Full Stack Deployment

- Deploying Node.js/Express to Render.com
- Railway.app for full-stack deployment
- MongoDB Atlas for cloud database hosting
- Environment configuration (production vs. development)
- Introduction to AWS/GCP/Azure cloud concepts
- Process management with PM2

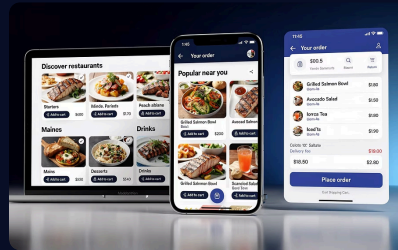
Industry-Level Real-World Projects

At WebiGeeks, learning by doing is our core philosophy. Every module culminates in hands-on practice, and the programme includes five industry-level capstone projects that mirror real client briefs. These projects become the centrepiece of your professional portfolio, demonstrating to employers that you can build production-grade applications – not just tutorials.



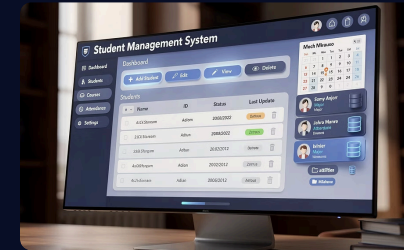
E-Commerce Platform

- Product listing with categories and search
- Shopping cart and wishlist system
- User authentication (register/login)
- Payment flow simulation
- Admin panel: manage products, orders, users



Food Ordering App

- Dynamic menu system with categories
- Cart and checkout with order summary
- Order management and tracking
- User authentication and profiles
- Restaurant admin dashboard



Student Management System

- Add, edit, and delete student records
- Analytics dashboard with charts
- Full database integration (MongoDB)
- Complete CRUD operations
- Search, sort, and filter functionality

Job Portal & Mini Projects

Job Portal System

A fully functional job portal that brings together employers and job-seekers on a single, authenticated platform.

- Job posting with rich descriptions and categories
- User registration and login (job-seeker & employer roles)
- Apply for jobs with resume upload
- Admin controls: approve/reject listings
- Application status tracking dashboard
- Search and filter jobs by location, skill, salary

Additional Portfolio Projects

These bite-sized but impactful projects demonstrate core JavaScript skills and make your GitHub profile shine.

- **Dice Game** – DOM manipulation, randomisation, two-player logic
- **Drum Kit** – Keyboard events, audio API, interactive UI
- **Blog Website** – EJS templating, CRUD, Express backend
- **Weather App** – Third-party API integration (OpenWeather), async/await
- **To-Do App** – Full CRUD with MongoDB, React frontend + Node backend

☐ All projects are added to your GitHub portfolio and reviewed by mentors before placement support begins.

Career Opportunities After Completing the Course

The WebiGeeks MERN Stack programme is designed with one goal in mind: your employment. Our curriculum is benchmarked against real job descriptions from top Indian and global tech companies. Graduates are equipped to apply for a wide range of high-demand roles in the software industry.



Full Stack Developer

Build end-to-end web applications handling both frontend and backend responsibilities. Average package: ₹4-12 LPA for freshers.



MERN Stack Developer

Specialist role using MongoDB, Express, React, and Node.js – one of the most in-demand job profiles in 2025-26.



Frontend Developer

Focused on building responsive, interactive user interfaces with React.js, CSS, and modern design systems.



Backend Developer

Build and maintain robust server-side systems, APIs, and databases using Node.js, Express, and MongoDB.

7

Months Duration

Intensive, structured programme

15+

Modules Covered

End-to-end curriculum

10+

Live Projects

Real portfolio work

100%

Placement Support

Dedicated career assistance

Why Choose WebiGeeks?

In a sea of online coding courses and bootcamps, WebiGeeks stands apart through its unwavering commitment to quality, industry relevance, and student outcomes. We don't just teach you to code – we prepare you to excel in a professional development environment from day one.



Live Industry Projects

Work on real-world, production-grade projects that go directly into your portfolio. Our projects are designed around actual client use cases – not contrived classroom exercises. Employers can see real evidence of your capabilities from the moment they review your GitHub.



Industry-Level Training

Our instructors are active software professionals, not just educators. The curriculum is continuously updated to match current industry demands, ensuring you learn the tools and workflows that companies are actually using in production today.



Dedicated Placement Support

Our career team provides résumé building, LinkedIn profile optimisation, mock technical interviews, and direct referrals to hiring partners. We stay with you until you land your first role – because your success is our success.



Flexible Online & Offline Modes

Learn the way that suits your lifestyle. Choose fully online sessions with recorded backups, or attend our in-person offline batches for a classroom experience with peer collaboration and face-to-face mentorship.

Technologies You Will Master

The WebiGeeks curriculum covers the complete modern web development technology stack – from the fundamental building blocks of the web to enterprise-grade frameworks and cloud deployment tools. Every technology listed below is a live, in-demand skill that employers actively seek.

HTML5

Semantic markup & web structure

CSS3

Styling, Flexbox & Grid

Bootstrap 5

Responsive UI framework

JavaScript ES6+

The language of the web

React.js

Modern frontend library

Node.js

Server-side JavaScript runtime

Express.js

Web application framework

MongoDB

NoSQL database & Atlas cloud

Mongoose

Elegant MongoDB object modelling

Git & GitHub

Version control & collaboration

TypeScript

Type-safe JavaScript at scale

Redux Toolkit

Modern React state management

REST APIs

RESTful API design & integration

JWT Auth

Secure authentication & authorisation

Deployment

Vercel, Render, Railway & cloud

ENROL NOW

Limited Seats Available

Begin Your Full Stack Journey with WebiGeeks Today

Don't let another batch fill up without you. WebiGeeks keeps batches intentionally small to ensure every student receives personalised mentorship, direct instructor access, and guaranteed placement support. Seats for the next cohort are filling fast – take the first step towards your new career right now.



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Available Monday–Saturday, 9 AM – 7 PM



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We respond within 24 hours



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Online & Offline batches available

"Your dream job as a Full Stack Developer is 7 months away. The only question is – will you start today?"

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